```
clc;
clear;
close all;

K = 10;
Nt = 100;
B = 20e6;
Pt = 10;
N0 = 1e-17;
Pn = N0 * B;

antenna_range = 20:20:300;
avgSINR = zeros(size(antenna_range));

for a = 1:length(antenna_range)

    Nt = antenna_range(a);

    H = (randn(K,Nt) + 1i*randn(K,Nt))/sqrt(2);

    W = H' * inv(H*H');

    W = W / norm(W,'fro') * sqrt(Pt);

    Heff = H * W;

    signal = abs(diag(Heff)).^2;

    interference = sum(abs(Heff).^2,2) - signal;

    SINR = signal ./ (interference + Pn);

    avgSINR(a) = mean(SINR);

end

figure;

plot(antenna_range,10*log10(avgSINR),'LineWidth',2);

grid on;

xlabel('Number of Antennas');
```

```
ylabel('Average SINR (dB)');

title('Single Cell Massive MIMO Downlink');
```

Full

```
clc;
clear;

K = 10;
N = 1e4;
B = 20e6;
N0 = 1e-17;

Pn = N0 * B;

antenna_range = 20:20:300;

Pt = 10;

avg_SINR_vs_Nt = zeros(size(antenna_range));

for a = 1:length(antenna_range)

    Nt = antenna_range(a);

    SINR_all = zeros(K, N);

    for iter = 1:N

        H = (randn(K, Nt) + 1i*randn(K, Nt)) / sqrt(2);

        W_zf = H' * inv(H * H');

        W_zf = W_zf / norm(W_zf,'fro') * sqrt(Pt);
```

```
        Heff = H * W_zf;

        signal_power = abs(diag(Heff)).^2;

        interference_power = sum(abs(Heff).^2,2) - signal_power;

        SINR = signal_power ./ (interference_power + Pn);

        SINR_all(:,iter) = SINR;

    end

    avg_SINR_vs_Nt(a) = mean(SINR_all(:));

end

figure;

plot(antenna_range,10*log10(avg_SINR_vs_Nt),'LineWidth',2);

grid on;

xlabel('Number of Transmit Antennas (N_t)');

ylabel('Average SINR (dB)');

title('SINR vs Number of Antennas (ZF Massive MIMO)');

Nt = 200;

Pt_dB_range = 0:2:30;

avg_throughput_vs_Pt = zeros(size(Pt_dB_range));

for p = 1:length(Pt_dB_range)

    Pt = 10^(Pt_dB_range(p)/10);

    throughput_all = zeros(K, N);

    for iter = 1:N

        H = (randn(K, Nt) + 1i*randn(K, Nt)) / sqrt(2);
```

```
        W_zf = H' * inv(H * H');
        W_zf = W_zf / norm(W_zf,'fro') * sqrt(Pt);
        Heff = H * W_zf;
        signal_power = abs(diag(Heff)).^2;
        interference_power = sum(abs(Heff).^2,2) - signal_power;
        SINR = signal_power ./ (interference_power + Pn);
        throughput = B * log2(1 + SINR);
        throughput_all(:,iter) = throughput;
    end
    avg_throughput_vs_Pt(p) = mean(throughput_all(:));
end
figure;
plot(Pt_dB_range, avg_throughput_vs_Pt/1e6,'LineWidth',2);
grid on;
xlabel('Transmit Power (dB)');
ylabel('Average Throughput (Mbps)');
title('Throughput vs Transmit Power (ZF Massive MIMO)');
```